FOUNDATIONS OF GENERATIVE AI (ICT4403)

Assignment 2: Autoregressive Text Generation System
Comparative Study of Simple RNN, Bidirectional RNN, LSTM, and GRU

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Assignment Title	Assignment 2 - Autoregressive Text Generation System
System Title	Comparative Word-Level Autoregressive Text Generation on Lewis Carroll's Alice in Wonderland using Simple RNN, Bidirectional RNN, LSTM and GRU
Dataset & Source	Alice's Adventures in Wonderland (Lewis Carroll) https://www.gutenberg.org/files/11/11-0.txt
GitHub Repository	https://github.com/sakshamgupta/autoregressive-text-generation-fgai

1. System Overview & Problem Statement

This project implements and evaluates an autoregressive next-word language generation system. Four distinct recurrent neural network architectures were built in PyTorch and evaluated under strictly comparable conditions on Lewis Carroll's *Alice's Adventures in Wonderland*: Simple RNN, Bidirectional RNN, LSTM, and GRU.
To eliminate data leakage, the corpus is split chronologically into 80% Train, 10% Validation, and 10% Test blocks, and the vocabulary is fitted solely on the training tokens. Input sequences of length 8 predict the succeeding token.

2. Quantitative Evaluation Summary (Best Epoch & Test Set)

Architecture	Parameters	Best Ep	Val Loss	Val PPL	Val Top-1	Test Loss	Test Top-1	Test Top-5
Simple RNN	985,022	3	5.0287	152.73	18.10%	5.1567	17.60%	38.41%
BiRNN	1,513,406	2	5.1401	170.74	16.88%	5.2474	16.94%	35.96%
LSTM	1,207,358	4	4.9778	145.16	18.70%	5.1261	**19.68%**	**39.17%**
GRU	1,133,246	4	**4.9701**	**144.04**	**18.92%**	5.1361	19.49%	38.95%

3. Cross-Model Temperature Analysis (Averaged Over Seeds)

Temperature	Distinct-1	Distinct-2	Trigram Repetition	Longest Loop	Empirical Quality
T = 0.2	0.507	0.701	18.7%	4–6 words	Grammatically strict; repetitive phrasing loops
T = 0.5	0.679	0.878	5.0%	0–4 words	Optimal empirical trade-off: high diversity with fluent phrasing
T = 1.0	0.855	1.000	0.0%	0 words	High lexical richness; slight syntactic drift across clauses
T = 1.5	0.905	1.000	0.0%	0 words	High entropy; ungrammatical, random token sequences

4. Conclusion & Model Comparison

• **LSTM and GRU** perform competitively as the top two architectures: GRU achieves the lowest validation loss (4.9701, PPL 144.04), while LSTM edges ahead on unseen test-set top-1 accuracy (19.68%) and top-5 accuracy (39.17%).
• **Bidirectional RNN:** By properly concatenating the forward final state h_n[-2] and backward final state h_n[-1], BiRNN captures the complete sequence context. However, with 1,513,406 parameters it exhibits earlier overfitting (peaking at epoch 2) compared to unidirectional models.
• **Sampling Strategy:** T = 0.5 provides the best empirical balance between lexical diversity (Distinct-1: ~0.68) and low phrase repetition (trigram repetition under 5%).